SEMESTER S5

DATA MINING (PECST525)

PROJECT ABSTRACT

Hybrid AI-Generated Image Detection

A Data Mining Approach to Digital Media Authentication

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PROJECT ABSTRACT

Hybrid AI-Generated Image Detection: A Data Mining Approach to Digital Media Authentication

This project applies core data mining techniques to solve the real-world classification problem of distinguishing AI-generated images from authentic photographs. Rather than treating this as a deep-learning or computer-vision task, the project frames it explicitly as a data mining pipeline — transforming unstructured image data into structured, mineable features, and then applying classical preprocessing, dimensionality reduction, classification, and clustering techniques to derive actionable insight.

The dataset used is the publicly available CIFAKE dataset, comprising real photographic images alongside AI-generated counterparts produced using diffusion-based models such as Stable Diffusion, providing a balanced and readily labeled foundation for supervised learning. This is further supplemented with additional AI-generated samples collected from other generative platforms (e.g., Midjourney, DALL-E) to improve generalizability across generator types. The images undergo data preprocessing: cleaning, resizing, and normalization to produce a consistent input format. From each image, a set of statistical and frequency-domain features is engineered, including noise residue patterns, FFT/DCT frequency-domain irregularities, color histogram statistics, and texture consistency measures, converting the unstructured image into a structured feature vector suitable for mining. Dimensionality reduction is then applied using Principal Component Analysis (PCA) to eliminate redundant and low-value attributes while retaining discriminative power.

For the classification stage, two supervised algorithms taught in the course are implemented and compared: a Decision Tree classifier built using Information Gain and Gini Index splitting criteria, and a Neural Network trained via backpropagation. Model performance is assessed using standard evaluation measures — accuracy, precision, recall, and F1-score — enabling a direct comparative analysis of the two approaches. Additionally, an unsupervised clustering stage using K-means is applied to the extracted feature space, examining whether images produced by different generative models form separable clusters, offering exploratory insight beyond the labeled classification task.

The project further incorporates a hybrid detection perspective by comparing the proposed forensic, feature-based approach against Google DeepMind's SynthID, an industry watermarking system that embeds imperceptible signals into AI-generated content at the point of generation. While SynthID offers high-confidence detection, its coverage is restricted to images produced by cooperating, watermark-enabled generators. The proposed data mining pipeline is positioned as a generalizable fallback capable of detecting AI-generated images irrespective of source model or watermark presence, and the two approaches are discussed as complementary layers of a practical detection system.

Through this project, the complete data mining lifecycle — data collection, preprocessing, feature engineering, dimensionality reduction, classification, evaluation, and clustering — is demonstrated end-to-end on a socially relevant, contemporary problem, fulfilling the course objective of applying data mining concepts and techniques to real-world application domains.

Dataset: *CIFAKE (Real vs. AI-Generated Images), supplemented with additional generator-specific samples*

Keywords: *Data Mining, Feature Extraction, Dimensionality Reduction, Decision Tree, Neural Network, K-means Clustering, Classification Evaluation, AI-Generated Image Detection, SynthID, Watermarking*